

# Shubhadeep Mandal

☎ 81602 63797   ✉ shubhadeepmandal11@gmail.com   📄 shubhadeep-mandal   🎧 Seronic2001

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>International Institute of Information Technology, Hyderabad</b><br><i>Telangana, India</i><br><i>Master of Technology in Computer Science and Engineering</i> | July 2025 – Present<br><br><i>CGPA: 9.12</i> |
| <b>Indian Institute of Information Technology, Ranchi</b><br><i>Ranchi, India</i><br><i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>    | Aug 2020 – Jun 2024<br><br><i>CGPA: 9.05</i> |

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
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| <b>Peer-to-Peer Distributed File Sharing System</b>   <i>C++, Sockets, Multithreading, OpenSSL</i>                                                                                                                                                                                                                                                                                                                                                                                                                    | 🔗 |
| <ul style="list-style-type: none"><li>– Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers.</li><li>– Engineered out-of-order parallel chunk writing via <code>pwrite()</code> and dynamic bitfield generation for simultaneous partial seeding.</li><li>– Architected Primary-Backup tracker failover with state synchronization, session flushing to prevent split-brain, and SHA-1 checks.</li></ul>                                   |   |
| <b>FOOL: POSIX-Compliant Interactive Shell</b>   <i>C++, POSIX APIs, Systems Programming, Linux</i>                                                                                                                                                                                                                                                                                                                                                                                                                   | 🔗 |
| <ul style="list-style-type: none"><li>– Engineered a modular UNIX shell in C++ supporting chained pipelines (<code>pipe</code>, <code>dup2</code>), quote parsing, and I/O redirection.</li><li>– Implemented job control to manage concurrent foreground and background tasks, preventing terminal deadlocks and zombie accumulation.</li><li>– Built an interactive terminal line editor with history navigation and non-blocking tab autocompletion for instant command discovery.</li></ul>                       |   |
| <b>AmbiStory: NLP Plausibility &amp; Disambiguation Engine</b>   <i>Python, PyTorch, Transformers, Embeddings</i>                                                                                                                                                                                                                                                                                                                                                                                                     | 🔗 |
| <ul style="list-style-type: none"><li>– Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching <math>\rho = 0.535</math> and 72.2% test accuracy.</li><li>– Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head.</li><li>– Boosted generalization by +0.265 <math>\rho</math> across unseen homonyms using group-stratified 5-fold CV, annotator KL divergence loss, and weight averaging.</li></ul> |   |
| <b>RookDB: In-House Relational Storage Engine</b>   <i>Rust, Storage Engine, Buffer Manager, Disk I/O</i>                                                                                                                                                                                                                                                                                                                                                                                                             | 🔗 |
| <ul style="list-style-type: none"><li>– Contributed to IIIT-H's in-house Rust RDBMS, architecting the type system, SQL-99 comparisons, and tuple memory layouts.</li><li>– Engineered explicit <code>RowLayout</code> structures with bit-packed null bitmaps, accelerating in-memory tuple deserialization.</li><li>– Integrated nullable row serialization with buffer-pool paging and authored benchmark suites profiling latency against PostgreSQL.</li></ul>                                                    |   |

## TECHNICAL SKILLS

**Languages:** C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.  
**Systems & OS:** POSIX APIs, Sockets, Linux/Unix, Multithreading, Git, Storage Engines.  
**AI, ML & NLP:** Transformers (DeBERTa, MPNet), PyTorch, Hugging Face, NLI, Sentence-Transformers, Scikit-learn.  
**Web Technologies:** Next.js, React, Node.js, Convex, Prisma, PostgreSQL, WebSockets.

## ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.